

## **AKTU Lucknow**

### **MID-TERM EXAMINATION ? 2025**

Concrete Technology ? Mid-Term Examination, 2025

Subject Code: CE302 Semester: 3 Regulation: R20

Time: 90 Minutes Max. Marks: 30

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.  
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#### **SECTION A (Unit 1: Cement and Aggregates)**

1. Explain Cement and Aggregates with a neat diagram/sketch and a suitable example. [12 Marks]
2. Solve a problem related to Cement and Aggregates. Show all the steps clearly. [12 Marks]
3. Describe the working principle of Cement and Aggregates with necessary equations. [8 Marks]

#### **SECTION B (Unit 2: Concrete Mix Design)**

4. Describe the working principle of Concrete Mix Design with necessary equations. [12 Marks]
5. Discuss the engineering applications of Concrete Mix Design. [12 Marks]
6. List the advantages and limitations of Concrete Mix Design. Explain any two in detail. [12 Marks]

#### **SECTION C (Unit 3: Fresh and Hardened Concrete)**

7. Solve a problem related to Fresh and Hardened Concrete. Show all the steps clearly. [12 Marks]
8. List the advantages and limitations of Fresh and Hardened Concrete. Explain any two in detail. [12 Marks]
9. Define Fresh and Hardened Concrete. Explain its significance in Concrete Technology with a suitable example. [12 Marks]

#### **SECTION D (Unit 4: Special Concretes)**

10. Compare and contrast the key concepts involved in Special Concretes. [8 Marks]
11. Solve a problem related to Special Concretes. Show all the steps clearly. [8 Marks]
12. Discuss the engineering applications of Special Concretes. [12 Marks]

#### **SECTION E (Unit 5: Quality Control and Testing)**

13. Discuss the engineering applications of Quality Control and Testing. [12 Marks]
14. List the advantages and limitations of Quality Control and Testing. Explain any two in detail. [12 Marks]

**15. Discuss the engineering applications of Quality Control and Testing. [12 Marks]**

\*\*\* End of Question Paper \*\*\*